Phase Shifted Full Bridge (PSFB) Converter-II

Introduction

In the rapidly evolving landscape of charging infrastructure and power electronics, the efficiency and reliability of power converters are paramount. The Phase Shifted Full Bridge (PSFB) converter stands out as a crucial topology, offering the advantage of Zero Voltage Switching (ZVS) to minimize switching losses and improve overall performance. This lecture delves into the intricate operational modes of the PSFB converter, building upon previous discussions to provide a comprehensive understanding of its behavior, particularly focusing on the mechanisms that enable soft switching and the inherent trade-offs involved. By dissecting each mode of operation, we uncover how ZVS is achieved for both the leading and lagging legs of the converter, and critically, how this soft-switching capability impacts the effective duty cycle and output voltage. Understanding these dynamics is essential for designing robust and efficient power conversion systems that meet the stringent demands of contemporary applications.

Core Concepts

Term	Definition	Significance
`Zero Voltage Switching (ZVS)`	A soft-switching technique where a power switch turns on when the voltage across it is zero, minimizing turn-on switching losses.	Crucial for improving converter efficiency, especially at high switching frequencies, by reducing energy dissipation during switching transitions.
`Leading Leg`	One half-bridge of the full-bridge converter (e.g., S1 and S2) where ZVS is typically easier to achieve due to the presence of the output inductor.	Its ZVS operation is generally robust across varying load conditions, contributing significantly to overall converter efficiency.
`Lagging Leg`	The other half-bridge of the full-bridge converter (e.g., S3 and S4) where ZVS is more	Its ZVS capability is critical for full soft-switching operation but often requires careful design considerations and may

Term	Definition	Significance
	challenging to achieve, especially at light loads.	be lost under certain conditions.
`Dead Time`	A short, intentional delay between the turn-off of one switch and the turn-on of its complementary switch in a half-bridge.	Essential for allowing resonant transitions to complete, ensuring ZVS, and preventing shoot-through currents that could damage switches.
`Duty Loss (D_loss)`	The reduction in the effective duty cycle of the converter due to the time required for soft-switching transitions.	Directly impacts the output voltage regulation, as the actual output voltage is proportional to (D - D_loss), necessitating compensation in control strategies.
`L_leakage`	The leakage inductance of the high-frequency transformer, which plays a vital role in the resonant transitions for ZVS.	Its value is critical for achieving ZVS, particularly for the lagging leg, but a larger value can increase duty loss.
`I_P`	The primary current flowing through the transformer, which reflects the load current and participates in resonant transitions.	Its magnitude and direction are key to determining the energy available for charging/discharging parasitic capacitances during ZVS.
`V_AB`	The voltage across the primary winding of the transformer, which is controlled by the switching actions of the full bridge.	Its waveform dictates the energy transfer to the secondary side and influences the resonant behavior during dead times.
`V_R`	The rectified voltage on the secondary side of the transformer, before the output filter.	Its behavior, especially during dead times, indicates the presence of freewheeling diodes and the effective duty cycle.

Detailed Analysis

The PSFB converter operates through a sequence of modes, each characterized by specific switch states, current paths, and voltage transitions. This lecture elaborates on the modes that facilitate Zero Voltage Switching (ZVS) and the resulting duty cycle loss.

1. PSFB Operating Modes (Recap and Continuation)

The lecture begins by recalling the initial modes and then proceeds to detail the subsequent stages of operation, particularly focusing on the transitions that enable ZVS for both the leading and lagging legs.

- **Mode 0 (t < t0):** This is the initial state where diagonal switches S1 and S4 are conducting. The primary voltage V_AB is V_in, and the rectified voltage V_R is also V_in (reflected to the primary). The primary current I_P is positive, and the secondary diode DR1 is conducting. Capacitors C1 and C2 are at 0V and V_in respectively, while C3 and C4 are at V_in and 0V.
- **Mode 1 (t0 < t < t1):** At t0, the gate pulse to S1 is removed. The primary current I_P, which is the reflected load current, now flows through the output capacitances C1 and C2 of the leading leg. C1 charges from 0V, and C2 discharges from V_in. V_AB, which is equal to V_C2, starts falling with a slope determined by I_P / (C1 + C2). DR1 continues to conduct.
- **Mode 2 (t1 < t < t2):** At t1, V_C2 reaches 0V, allowing the body diode D2 of switch S2 to turn on. This creates a zero-voltage condition across S2. The dead time between S1 turning off and S2 turning on must be greater than T01 (the duration of Mode 1) to ensure S2 achieves ZVS turn-on. During this mode, S4 and D2 are conducting, and DR1 is still conducting.
- **Mode 3 (t2 < t < t3):** At t2, S2 is turned on with ZVS. Now, the leading leg has switched from S1 to S2. At t2, the gate pulse to S4 (lagging leg) is removed. The primary current I_P now flows through the output capacitances C3 and C4 of the lagging leg. V_AB becomes -V_C4. As C4 charges and C3 discharges, V_AB starts to rise towards -V_in. During this transition, the primary current I_P is no longer sufficient to maintain the reflected load current, causing both secondary diodes DR1 and DR2 to become forward biased. This effectively shorts the secondary winding, making V_R zero. The L_leakage resonates with C3 and C4.
- **Mode 4 (t3 < t < t4):** At t3, V_C3 reaches 0V, allowing the body diode D3 of switch S3 to turn on. This creates a zero-voltage condition across S3. The dead time for the lagging leg must be greater than T23 (the duration of Mode 3) to ensure S3 achieves ZVS turn-on. During this mode, S2 and D3

are conducting, and both DR1 and DR2 are forward biased, keeping V_R at zero. V_AB is now -V_in. The primary current I_P starts falling linearly with a slope of -V_in / L_leakage.

- **Mode 5 (t4 < t < t5):** At t4, the primary current I_P reaches zero and then continues to flow in the negative direction. S2 and S3 remain on, and DR1 and DR2 are still forward biased. V_AB remains at -V_in. I_P continues to fall linearly until it reaches the reflected negative load current (-n * I_L).
- **Mode 6 (t > t5):** At t5, I_P reaches -n * I_L. At this point, DR1 becomes reverse biased, and DR2 takes over the entire load current. V_R transitions from zero to n * V_in (reflected to the primary). The converter is now in a state where S2 and S3 are conducting, and DR2 is conducting. This completes one half-cycle of operation, and the process repeats symmetrically for the next half-cycle with S2 and S3 turning off, and S1 and S4 turning on.

2. Zero Voltage Switching (ZVS) Conditions

Achieving ZVS is critical for efficiency, but it requires specific conditions to be met. These conditions are categorized as "required" (related to timing) and "necessary" (related to energy).

- **Required Conditions for ZVS:**
- **Leading Leg:** The dead time provided for the leading leg switches (e.g., between S1 turn-off and S2 turn-on) must be greater than T01. This duration is approximately 2 * C * V_in / I1, where C is the output capacitance and I1 is the primary current.
- **Lagging Leg:** The dead time for the lagging leg switches (e.g., between S4 turn-off and S3 turn-on) must be greater than T23. This duration is given by $1/\omega 1 * \sin^{-1}(V_{in} / I2 * \sqrt{(2C / L_{leakage})})$, where ω1 is the resonant frequency, I2 is the primary current, and C is the output capacitance.
- **Necessary Conditions for ZVS (Energy Balance):**
- **Leading Leg:** The energy stored in the primary-side inductance ($L_{leakage} + L/N^2$) at the moment of switch turn-off must be sufficient to charge and discharge the output capacitances of the leading leg. Specifically, $0.5 * (L_{leakage} + L/N^2) * I1^2$ must be greater than $0.5 * C1 * V_{in}^2 + 0.5 * C2 * V_{in}^2$. The presence of the output inductor (L) reflected to the primary side makes ZVS easier to achieve for the leading leg, as it provides a larger inductance and thus more stored energy.
- **Lagging Leg:** For the lagging leg, only the leakage inductance (L_leakage) is involved in the resonant transition. The energy stored in L_leakage must

be sufficient to charge and discharge the lagging leg capacitances: $0.5 * L_{leakage} * I_P^2$ must be greater than $0.5 * C3 * V_{in}^2 + 0.5 * C4 * V_{in}^2$. This condition is more challenging to meet, especially under light load conditions where I_P is small, or if $L_{leakage}$ is small, or if C3/C4 are large, or at high input voltages. This makes the lagging leg more susceptible to losing ZVS.

3. Duty Cycle Loss (D_{loss})

While ZVS significantly reduces switching losses, it comes at the cost of a reduction in the effective duty cycle, known as duty loss.

- **Origin of Duty Loss:** The duty loss occurs during the period when the secondary winding is effectively short-circuited ($V_R = 0$), which happens from T2 to T5. During this time, no power is transferred to the output, effectively reducing the duration for which V_{in} is applied across the primary.
- **Calculation of Duty Loss:** The total time duration of duty loss in one half-cycle is $T25 = T23 + T34 + T45$. The duty loss (D_{loss}) is then calculated as T25 divided by the half-period ($T_S / 2$).
- An approximation can be made by considering T23 to be very small due to the high resonant frequency of $L_{leakage}$ with small output capacitances. Thus, D_{loss} is primarily determined by T34 and T45.
- The expression for D_{loss} can be derived as:
  $D_{loss} = (L_{leakage} * I_P(T3) / V_{in} + L_{leakage} * I_P(T5) / V_{in}) / (T_S / 2)$
  Assuming $I_P(T3) \approx I_P(T5) \approx n * I0$ (reflected output current), the simplified expression becomes:
  $D_{loss} = 4 * L_{leakage} * n * I0 * F_{SW} / V_{in}$
  where n is the transformer turns ratio, I0 is the output current, and F_{SW} is the switching frequency.
- **Implications of Duty Loss:**
- **Compromise:** To achieve ZVS, especially for the lagging leg, a larger $L_{leakage}$ might be desired. However, a larger $L_{leakage}$ directly increases D_{loss}, leading to a reduction in the effective duty cycle and thus the output voltage ($V0 = n * V_{in} * (D - D_{loss})$).
- **Load Dependence:** D_{loss} increases with higher output current (I0), meaning the duty loss is more significant at heavier loads.
- **Input Voltage Dependence:** D_{loss} decreases with increasing input voltage (V_{in}).

- **Switching Frequency Dependence:** D_loss increases with higher switching frequency (F_SW).
- **Impact on Output Voltage:** The actual output voltage of the PSFB converter is reduced by the duty loss, meaning V0 = N * V_in * (D - D_loss). This necessitates careful design and control to compensate for this inherent loss in the effective duty cycle.

Key Takeaways

- [ ] PSFB converters utilize Zero Voltage Switching (ZVS) to reduce switching losses and improve efficiency.
- [ ] ZVS is achieved by allowing resonant transitions during dead times, charging and discharging parasitic capacitances.
- [ ] The leading leg (e.g., S1/S2) typically achieves ZVS more easily due to the larger effective inductance (L_leakage + reflected output inductor).
- [ ] The lagging leg (e.g., S3/S4) ZVS is more challenging, especially at light loads, as only L_leakage is involved in the resonant transition.
- [ ] Dead time duration must be carefully chosen to be greater than the ZVS transition time (T01 for leading, T23 for lagging).
- [ ] Necessary ZVS conditions involve sufficient energy stored in inductances to charge/discharge switch capacitances.
- [ ] Implementing ZVS leads to a "duty loss" (D_loss) because the secondary winding is short-circuited during resonant transitions, preventing power transfer.
- [ ] Duty loss increases with higher leakage inductance, output current, and switching frequency, but decreases with higher input voltage.
- [ ] The actual output voltage is reduced by the duty loss, impacting the converter's effective duty cycle.
- [ ] There is a design trade-off between ensuring robust ZVS (potentially requiring larger L_leakage) and minimizing duty loss.

Conclusion

The detailed analysis of the Phase Shifted Full Bridge converter's operating modes reveals the intricate dance between switching actions, resonant transitions, and power delivery. While ZVS offers a compelling advantage in reducing switching losses, particularly crucial for high-frequency operation in applications like electric vehicle charging and renewable energy systems, its implementation introduces a quantifiable duty loss. This loss, stemming from the necessary dead times and resonant periods, directly impacts the converter's effective duty cycle and,

consequently, its output voltage. Designers must carefully balance the need for robust ZVS, especially for the more challenging lagging leg, against the desire to minimize duty loss. Understanding the factors influencing duty loss—such as leakage inductance, load current, input voltage, and switching frequency—is paramount for optimizing PSFB converter performance. Future discussions will build upon this foundation, exploring closed-loop control strategies to mitigate the effects of duty loss and further enhance the practical utility of PSFB converters in real-world power conversion scenarios.